

1.⑤ $\xi = y, \eta = x^3 - y. u_{\xi\eta} + u_{\xi} = 0. u = f(\xi)e^{-\eta} + g(\eta).$
 $u(x, y) = [y - g(0)]e^{-(x^3 - y)} + [3(x^3 - y) + g(0)e^{-(x^3 - y)}] = ye^{y - x^3} + 3(x^3 - y).$

2.④ $u(x, t) = t^2 + v(x, t), v(x, t) = (x + t)^2 + 2(x + t) + \begin{cases} -2(x - t), & x - t \geq 0; \\ e^{-2(x - t)} - 2(x - t)^2 - 1, & x - t \leq 0. \end{cases}$

3.⑦ $u = v + 2xt, v(x, t) = \sum_{k=0}^{\infty} T_k(t) \cos((2k + 1)x), x - \frac{\pi}{2} = \sum_{k=0}^{\infty} a_k \cos \lambda_k x,$
 $a_k = -\frac{4}{\pi \lambda_k^2}, \lambda_k = 2k + 1, k = 0, 1, 2, \dots;$
 $T_0''(t) - 9T_0(t) = 36e^{-3t}, T_0(0) = 0, T_0'(0) = -6 \Rightarrow T_0(t) = -6te^{-3t}.$
 $k = 1, 2, \dots: T_k''(t) + \gamma_k^2 T_k(t) = -9\pi a_k e^{-3t}, T_k(0) = T_k'(0) = 0, \text{ где}$
 $\gamma_k^2 = 2\lambda_k^2 - 11 = 2(2k + 1)^2 - 11, \Rightarrow T_k(t) = \frac{9\pi a_k}{9 + \gamma_k^2} \cdot \left[\cos(\gamma_k t) - \frac{3}{\gamma_k} \sin(\gamma_k t) - e^{-3t} \right].$

4.③ $u = r^3 \cos 2\varphi + \left(r^2 + \frac{4}{r^2}\right) \cos 2\varphi + \left(r + \frac{4}{r}\right) \sin \varphi.$

5.④ $u = (5e^{-5t} - 5 \cos t - \sin t) \cdot \sin(x + 2y) + e^{-2t} x \sin(x - z) + 2te^{-2t} \cos(x - z).$

6.⑥ $u(x) = 4\lambda C_1 \sin x + 4\lambda C_2 \cos x + x \sin(x - \alpha), x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$
 где $C_1 = \int_{-\pi/2}^{\pi/2} \cos y \cdot u(y) dy, C_2 = \int_{-\pi/2}^{\pi/2} \sin y \cdot u(y) dy, \begin{cases} C_1 - 2\pi\lambda C_2 = \frac{\pi}{4} \cos \alpha, \\ -2\pi\lambda C_1 + C_2 = -\frac{\pi}{4} \sin \alpha. \end{cases}$

1) $\lambda \neq \pm \frac{1}{2\pi}$: решение существует при любом $\alpha \in \mathbb{R}$ и

$$u(x) = \frac{\pi\lambda}{1 - 4\pi^2\lambda^2} [(\cos \alpha - 2\pi\lambda \sin \alpha) \sin x + (-\sin \alpha + 2\pi\lambda \cos \alpha) \cos x] + x \sin(x - \alpha),$$

$$x \in \left[-\frac{\pi}{2}; \frac{\pi}{2}\right];$$

2) $\lambda = \frac{1}{2\pi}$. Условие разрешимости $\cos \alpha = \sin \alpha$:

$$u(x) = C(\cos x + \sin x) + \frac{1}{2} \sin \alpha \sin x + x \sin(x - \alpha) = D(\cos x + \sin x) - \frac{1}{2} \sin \alpha \cos x +$$

$$+ x \sin(x - \alpha), x \in \left[-\frac{\pi}{2}; \frac{\pi}{2}\right], C \in \mathbb{R}, D \in \mathbb{R};$$

3) $\lambda = -\frac{1}{2\pi}$. Условие разрешимости $\cos \alpha = -\sin \alpha$:

$$u(x) = C(\sin x - \cos x) + \frac{1}{2} \sin \alpha \cos x + x \sin(x - \alpha) = D(\sin x - \cos x) + \frac{1}{2} \sin \alpha \sin x +$$

$$+ x \sin(x - \alpha), x \in \left[-\frac{\pi}{2}; \frac{\pi}{2}\right], C \in \mathbb{R}, D \in \mathbb{R};$$

4) $u_1(x) = \sin x + \cos x, \lambda_1 = \frac{1}{2\pi}; u_2(x) = \sin x - \cos x, \lambda_2 = -\frac{1}{2\pi}.$

7.④ $u(r, \varphi, t) = \sum_{k=2}^{\infty} a_k e^{-(\mu_k^{(1)})^2 t} \cdot J_1(\mu_k^{(1)} r) \cos \varphi +$
 $+ \left[a_1 - \frac{(\mu_1^{(1)})^2}{1 + (\mu_1^{(1)})^4} e^{-(\mu_1^{(1)})^2 t} + \frac{1}{1 + (\mu_1^{(1)})^4} \left((\mu_1^{(1)})^2 \cos t + \sin t \right) \right] \cdot J_1(\mu_1^{(1)} r) \cos \varphi,$
 где $a_k = \frac{\int_0^1 J_2(\mu_1^{(2)} r) J_1(\mu_k^{(1)} r) r dr}{\int_0^1 (J_1(\mu_k^{(1)} r))^2 r dr}, k = 1, 2, \dots.$